

LAB 3 ARCHITECTURAL JUSTIFICATION

Customizable Subscription Box Scheduler | Rhea Menon, PES1UG24AM222 | 9/10/2026

Architecture Selection

We chose Microservices Architecture for the Customizable Subscription Box Scheduler.

Architectural Choice

The system is split into independent services: a Subscription & Preference Service (preference tags, box swap/pause, skip-cycle - FR-001-003), a Billing & Payment Service (automated monthly charging - FR-004, talking to the external Payment Gateway), and a Fulfillment Manifest Service (manifest generation/export - FR-005, NFR-001), each reached through its own portal (Subscriber Portal, Fulfillment Console) and its own interfaces, with a shared Subscriber Data Store for persistence.

Two Reasons for This Choice

- NFR-001 requires the fulfillment manifest generator to export shipping labels for 10,000 monthly boxes in under a minute - a bursty, batch workload that only spikes around the shipping window. As its own microservice, the Fulfillment Manifest Service can be scaled independently for that window without over-provisioning the Subscriber Portal or Subscription & Preference Service, which handle steady, low-latency, real-time requests the rest of the month.
- The system has clearly separate bounded contexts owned by different actors: Subscriber-facing customization (FR-001-003), automated billing (FR-004), and Fulfillment Lead-driven manifest export (FR-005). Microservices lets the Billing & Payment Service be developed, deployed, and audited under strict PCI-DSS controls on

Security Advantage

Payment handling is isolated in its own Billing & Payment Service, reachable only through the IBillingEligibility and IPaymentGatewayAPI interfaces, so raw payment data is never stored in or passed through the Subscriber Portal or Subscription & Preference Service. This confines the PCI-DSS compliance boundary from NFR-002 to a single, independently secured service instead of the whole system, and keeps encrypted payment storage in one auditable place.

Performance Benefit

Because the Fulfillment Manifest Service is a separate microservice, it can be scaled out with extra instances specifically during the shipping window to meet NFR-001's under-a-minute target for 10,000 boxes - without needing to over-provision the Subscriber Portal or Subscription & Preference Service, which see a completely different, low-volume-per-request interactive traffic pattern the rest of the time.